OpenID Connect For .NET Developers

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MVP Microsoft Most Valuable Professional

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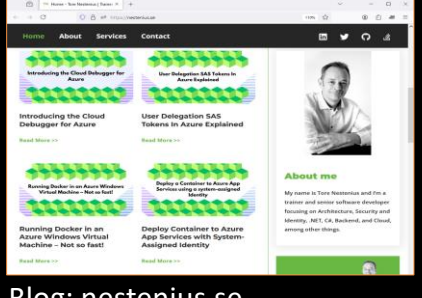
1

About Tore Nestenius

Work: tn-data.se



Blog: nestenius.se

meetup.com/net-skane

Have a question about this session or beyond?

tore@tn-data.se

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2

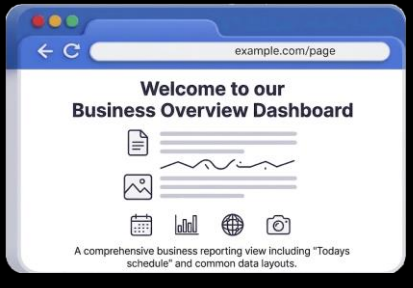
First, One Important Thing

Do Ask Questions!

The New Project

The New Project

You Just Built This Awesome ASP.NET Core Web App



What Did We Forget?

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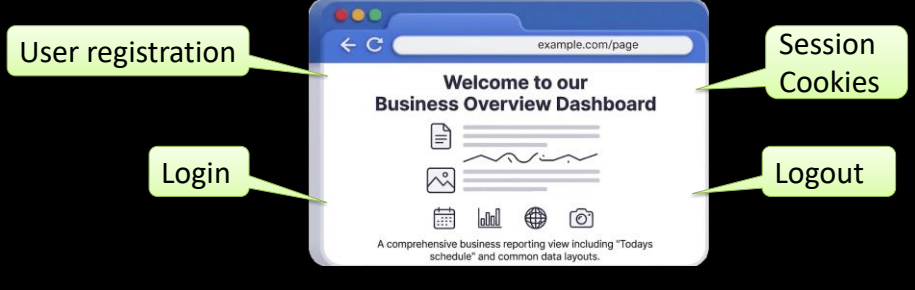
5

The New Project

Your Client Asks You To Add Authentication!



But It Gets Worse...

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6

The New Project

The Client Has More Requirements

So, How Do We Support This?

7

We Need To Use OpenID Connect!

OpenID®

But How Many of Us Can Explain It?

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8

OpenID Connect

Example: I Have My Photos in Google Photos

You

Google Photos

I Want a Photo Printing Service to Print Them

OpenID Connect

I Need to Give the Photo Printer Access to My Photos

You

Photo Printer

Google Photos

How Do I Give It Access?

The Photo Printer Needs an Access Token

How Does the Photo Printer Get the Token?

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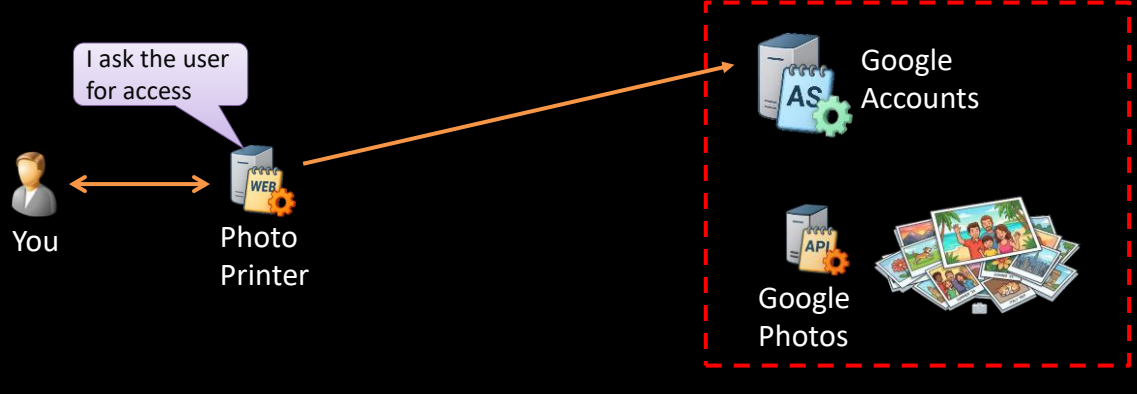
11

It Asks Google Accounts for a Token



What Happens Next?

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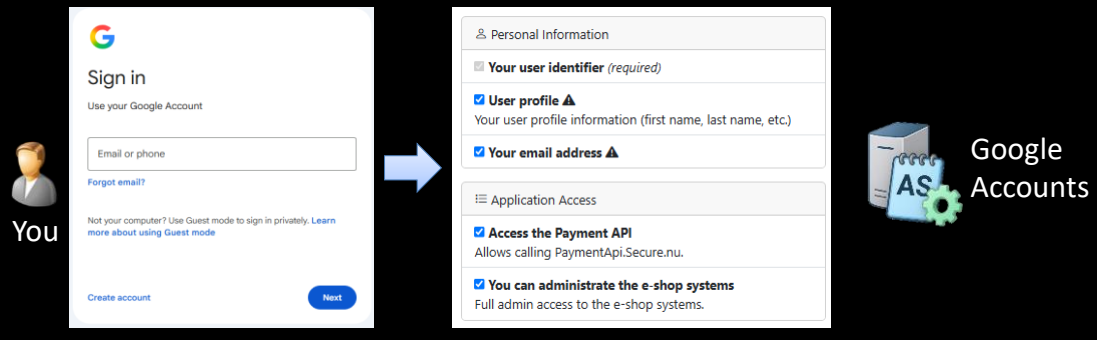
12

OpenID Connect

The User Authenticate and Give Consent



What Happens Next?

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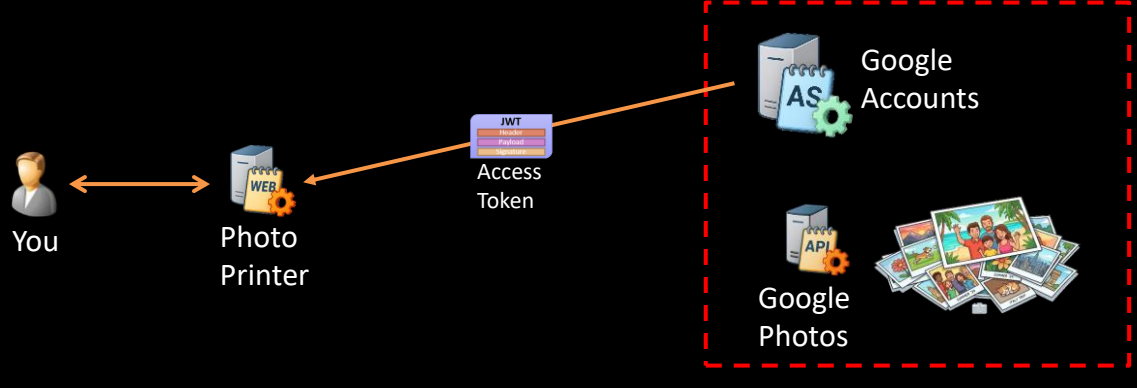
13

OpenID Connect

The Tokens Are Returned To the Client



How Does the Photo Printer Use the Token?

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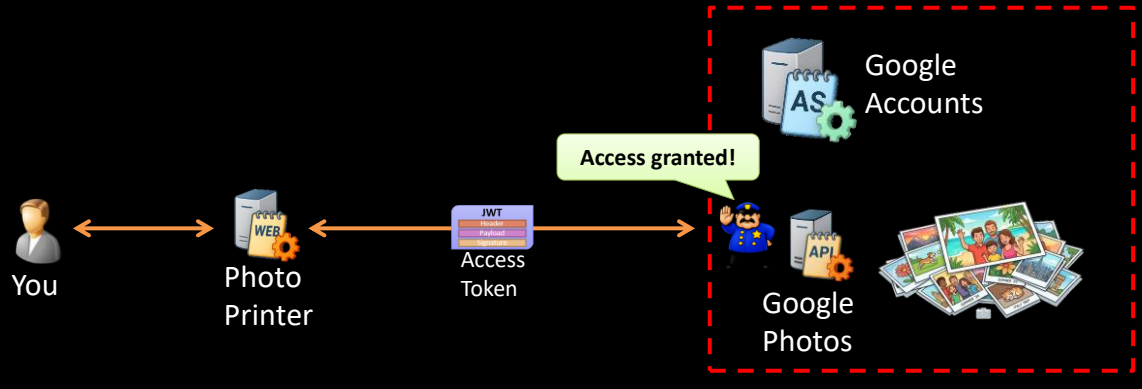
14

OpenID Connect

The Photo Printer Uses the Token to Access My Photos



Let us build it in ASP.NET Core

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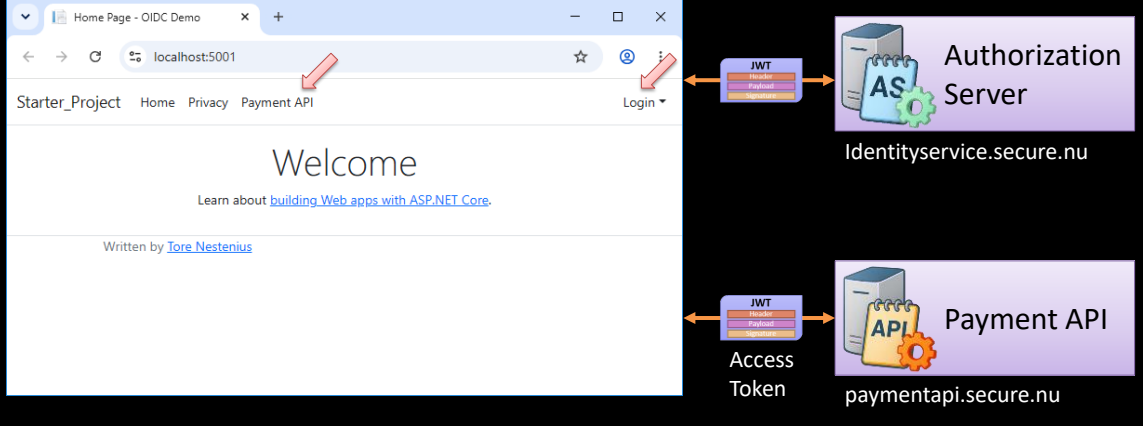
15

Demo #1 – Introducing the Demo Application

Client Application



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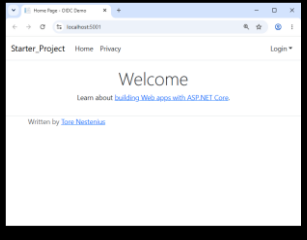
16

The Needs of the Client



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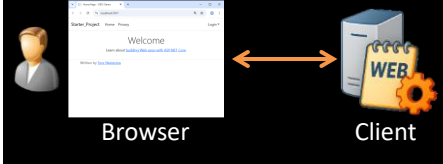
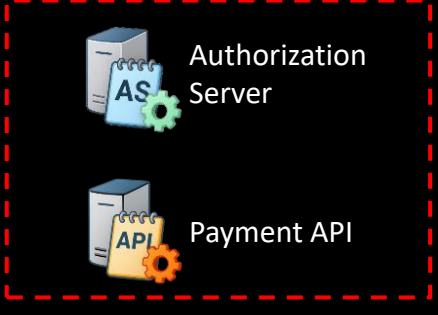
17

The Needs Of The Client

Here is a Typical Setup





What Does a Client Actually Want?

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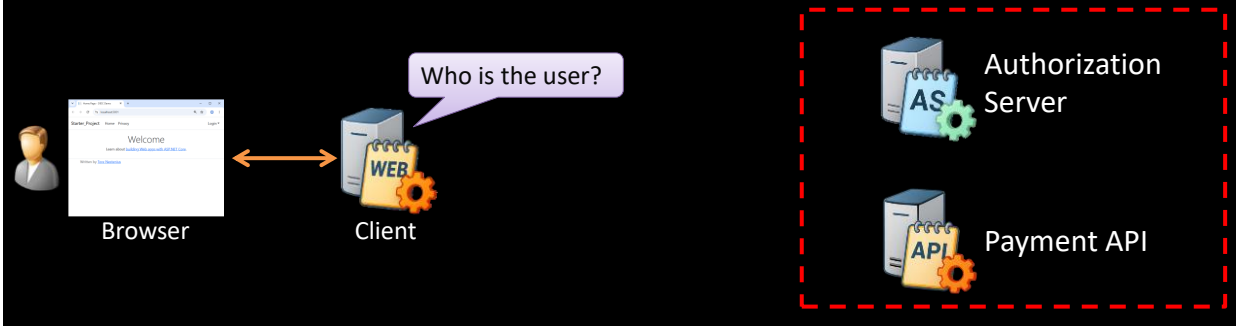
18

The Needs Of The Client

First, It Needs to Know Who the User Is



How Does The Client Get This Information?

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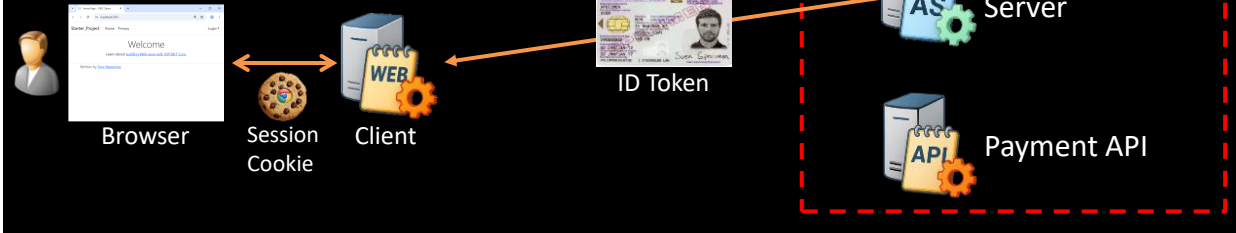
19

The Needs Of The Client

The Client Gets an ID Token That Identifies the User



But the Client Also Has Another Need

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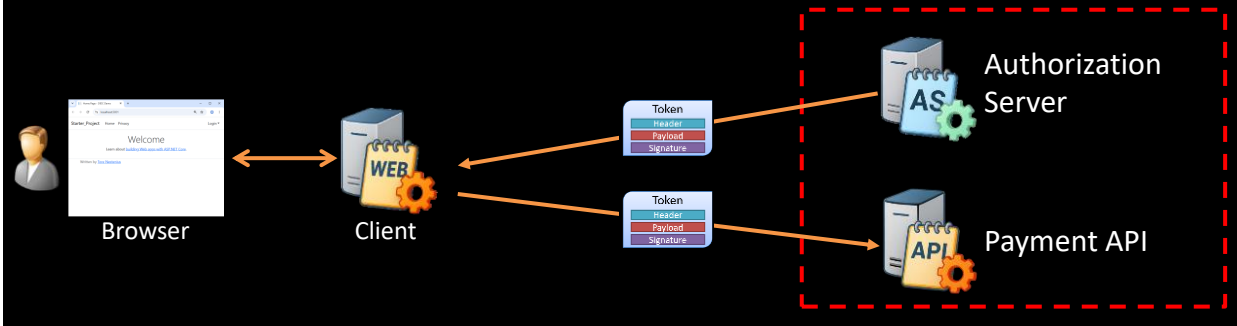
20

The Needs Of The Client

Second, It Needs to Access the Payment API



The Client Uses the Access Token to Call the API

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21

ASP.NET Core Request Pipeline

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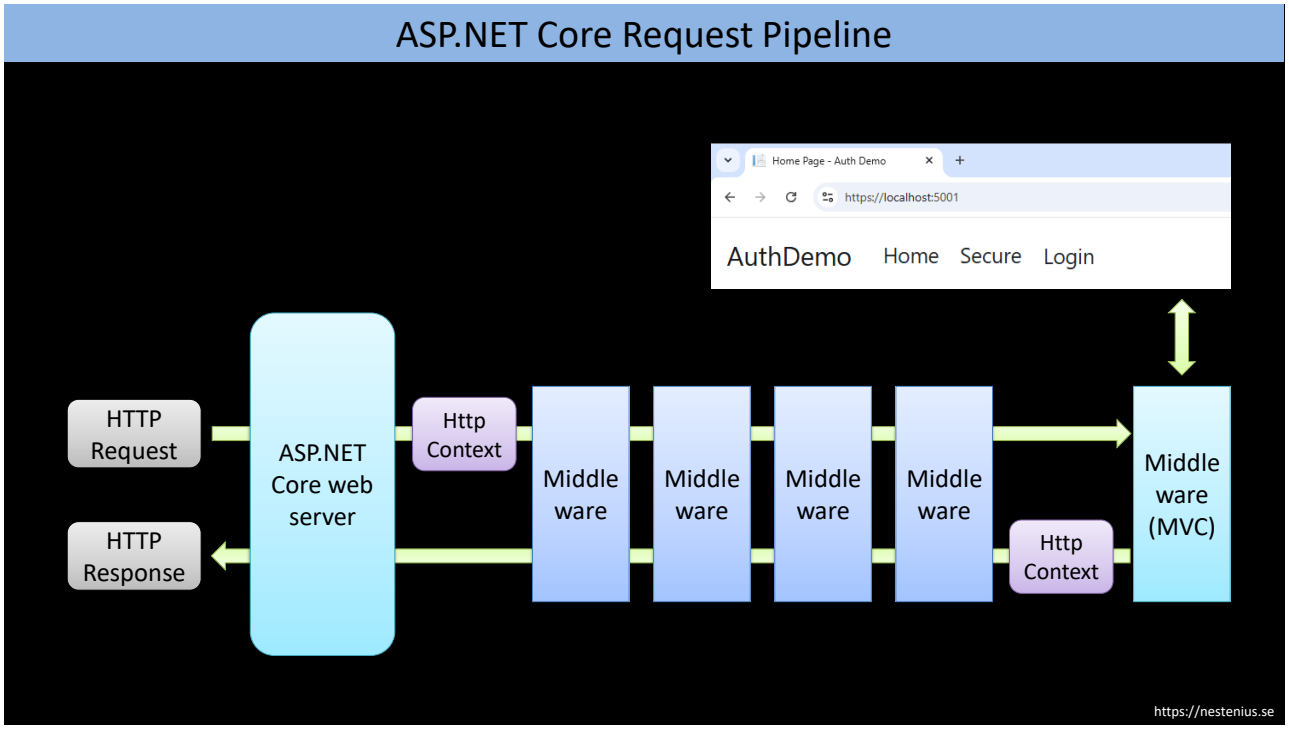
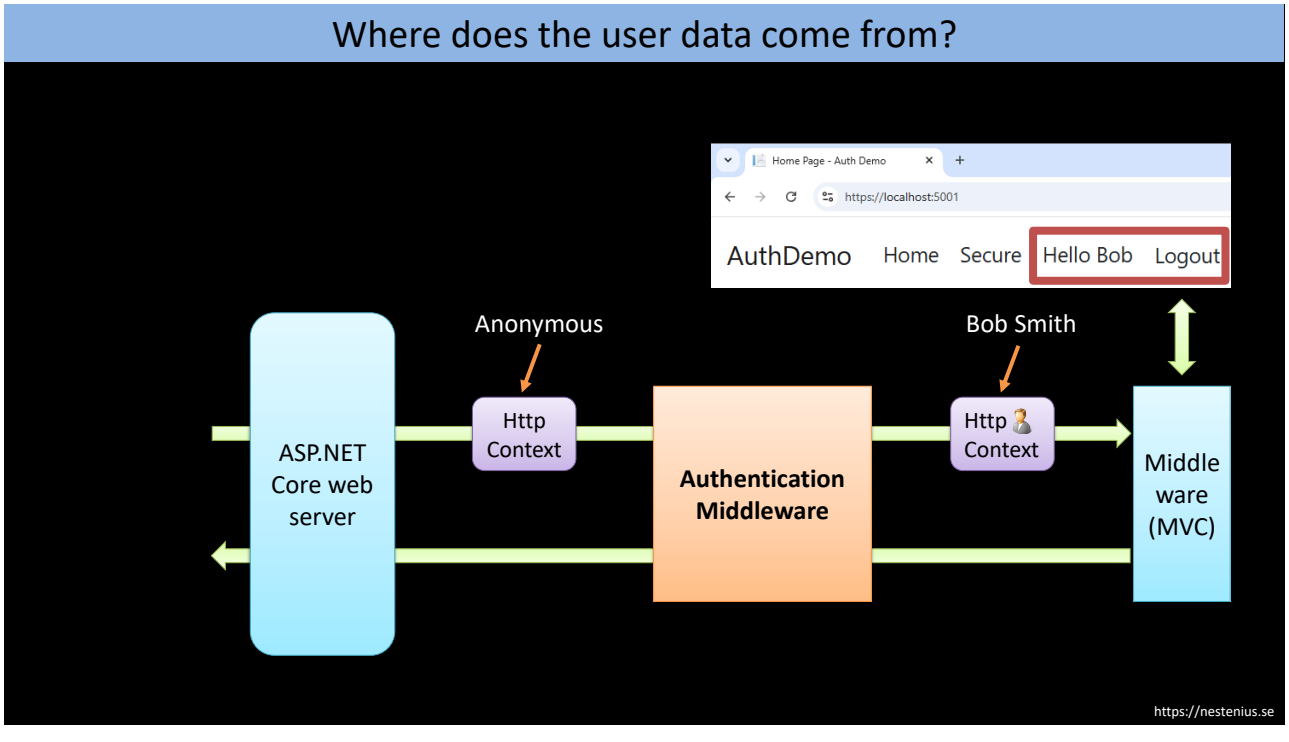
22

ASP.NET Core Request Pipeline



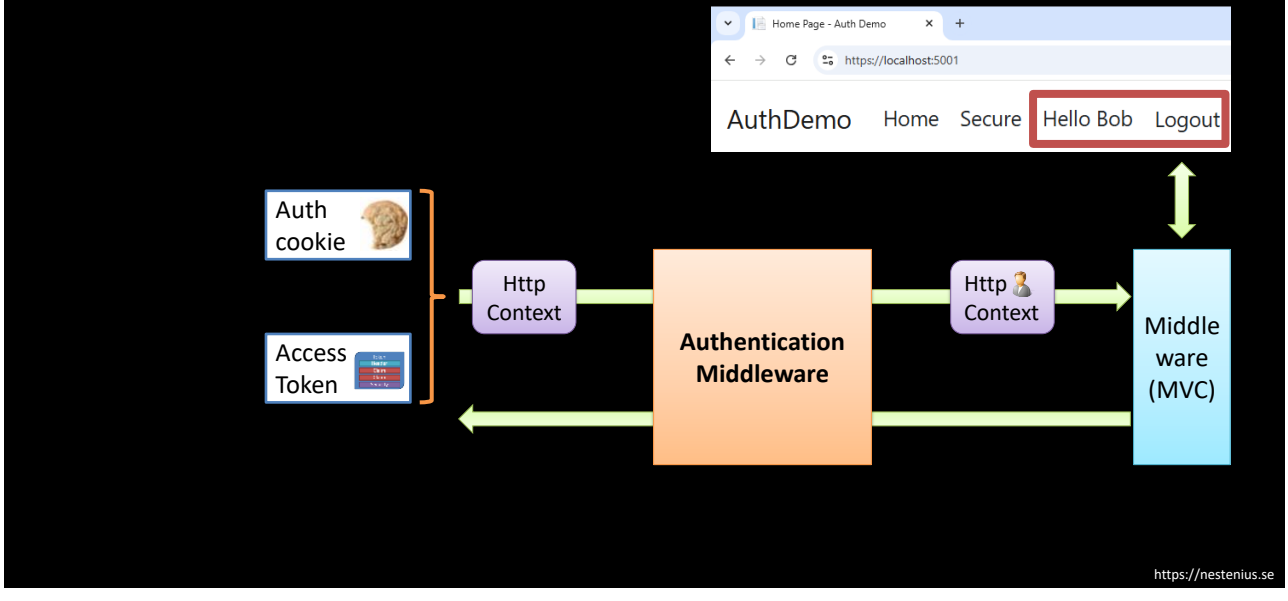
23

Where does the user data come from?



24

Where does the user data come from?



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25

Live coding #2 – Adding The Middleware

```
builder.Services.AddAuthentication();

...

app.UseAuthentication();
```

```
public async Task Login()
{
    var prop = new AuthenticationProperties
    {
        RedirectUri = "/"
    };


    // Ask the user to sign in
    await HttpContext.ChallengeAsync("oidc", prop);
}
```

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26

Adding OpenID Connect

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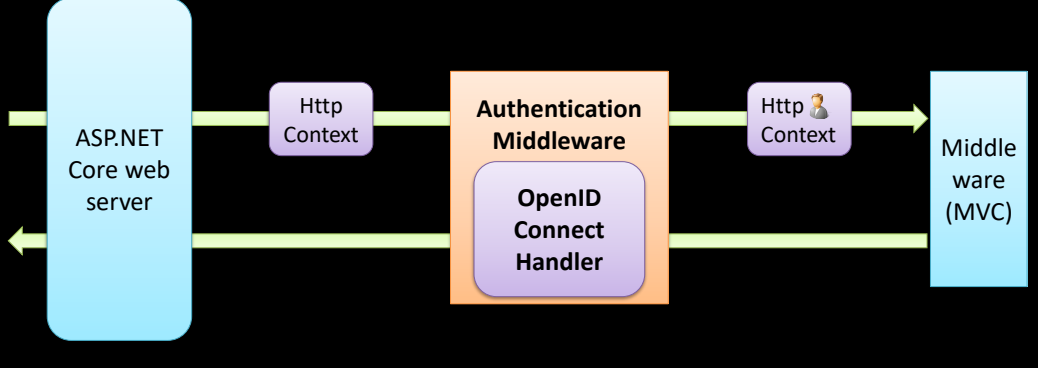
Adding OpenID Connect

We Start by Adding the OpenID Connect Handler



What Is the Handler's Main Purpose?

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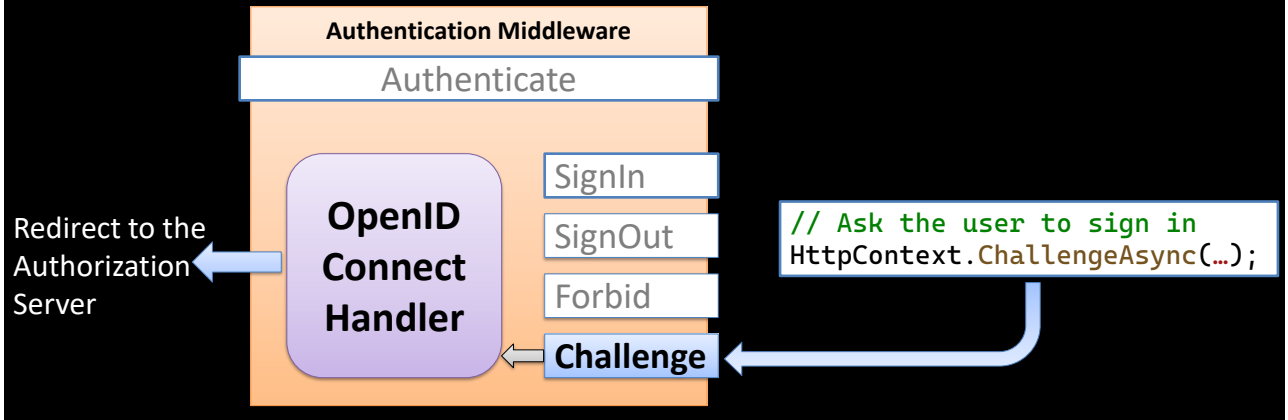
28

Adding OpenID Connect

It Should Start the Authentication Flow



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29

Live coding #3

```
builder.Services.AddAuthentication(o =>
{
    o.DefaultScheme = "cookie";
    o.DefaultChallengeScheme = "oidc";
})
.AddOpenIdConnect("oidc", o =>
{
    o.Authority = "https://identityservice.secure.nu";
    o.ResponseType = "code";

    o.ClientId = "localhost-addoidc-client";
    o.ClientSecret = "mysecret";

    o.Scope.Add("openid");
    o.Scope.Add("email");
    o.Scope.Add("profile");
    o.Scope.Add("payment");

    o.MapInboundClaims = false;
    o.Prompt = "login";
    o.PushedAuthorizationBehavior = PushedAuthorizationBehavior.Disable;
});
```

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30

Authorization Code Flow

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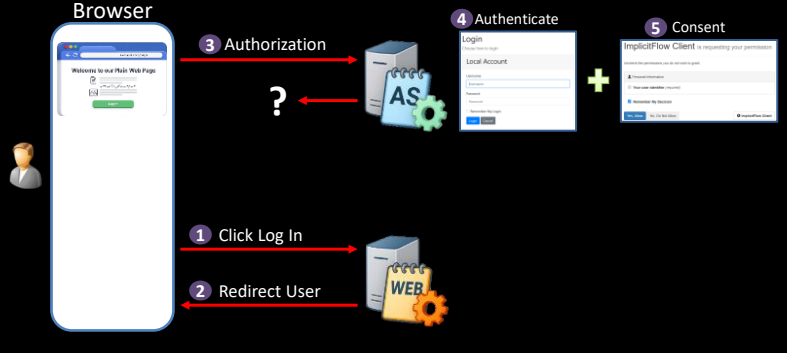
31

Authorization Code Flow

The User Is Redirected to Sign In and Give Consent



What Is Sent Back After Consent?

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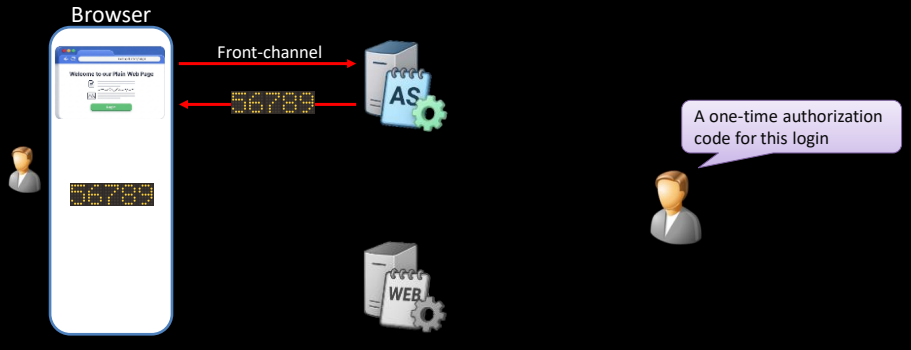
32

Authorization Code Flow

An Authorization Code Is Returned



What Happens Next?

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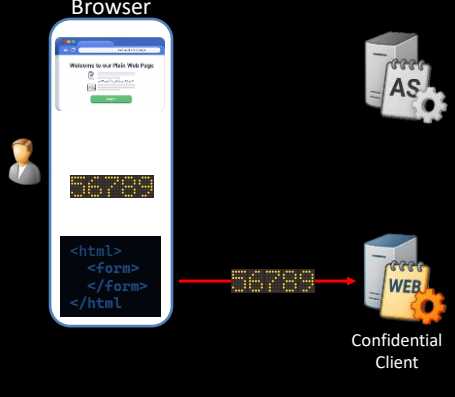
33

Authorization Code Flow

The Browser Forwards the Code to the Client



What Does the Client Do With It?

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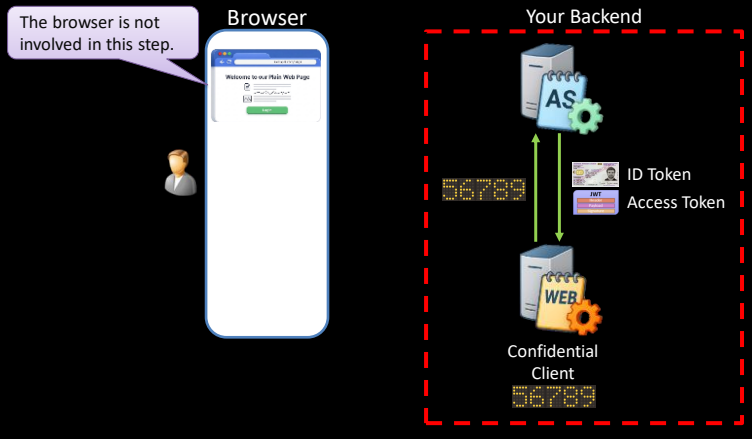
34

Authorization Code Flow

The Client Exchanges The Code For Tokens



What Happens To The Tokens?

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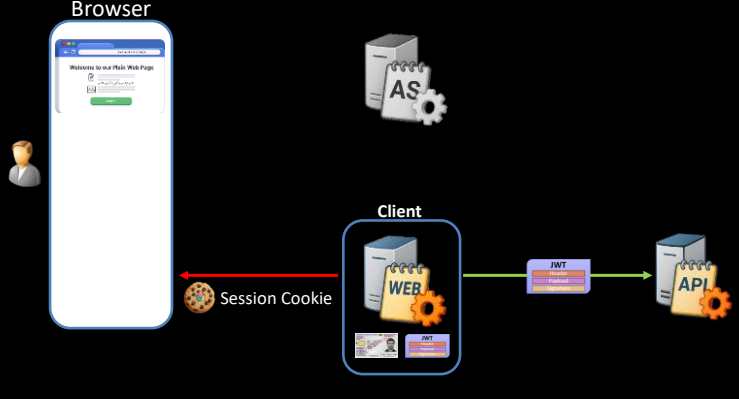
35

Authorization Code Flow

The Client Uses the Tokens



Let us Explore This In Our Demo

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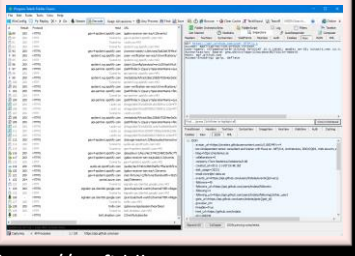
36

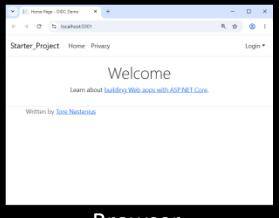
Live coding #4

Authorization Server

Fiddler Classic



https://getfiddler.com



Browser

Client

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37

The Missing Piece

An unhandled exception occurred while processing the request.

InvalidOperationException: No sign-in authentication handlers are registered. Did you forget to call AddAuthentication().AddCookie("cookie",...)?

Microsoft.AspNetCore.Authentication.AuthenticationService.SignInAsync(HttpContext context, string scheme, ClaimsPrincipal principal, AuthenticationProperties properties)

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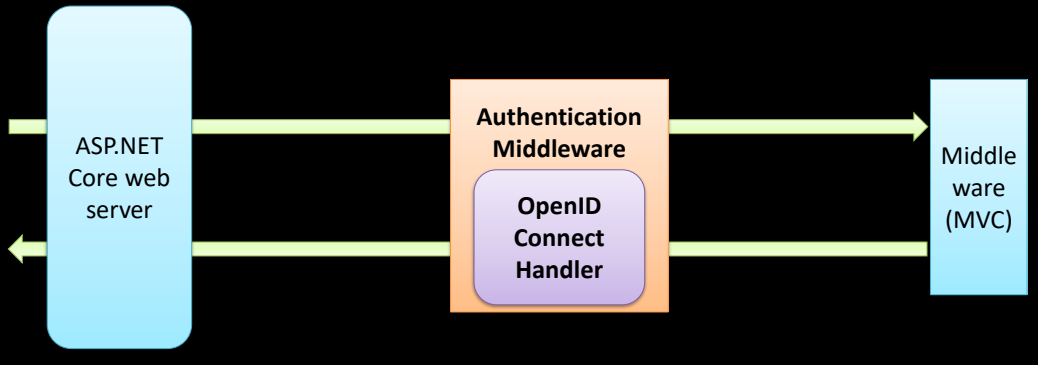
38

The Missing Piece

So Far, Our Application Looks like This



What Has the Handler Done So Far?

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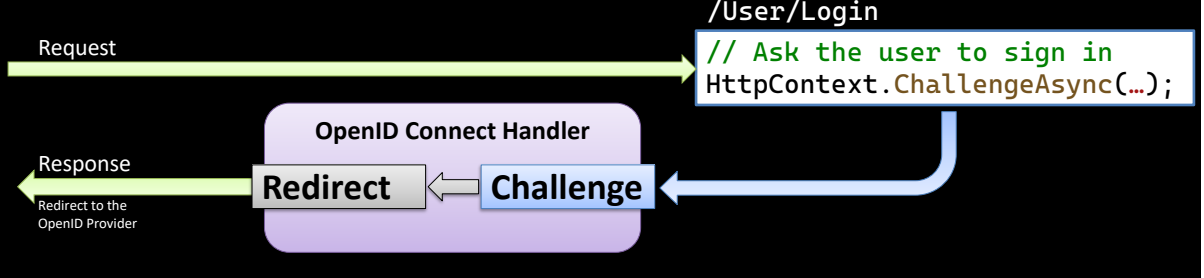
39

The Missing Piece

Handle the Authentication Challenge



What Else Is It Responsible For?

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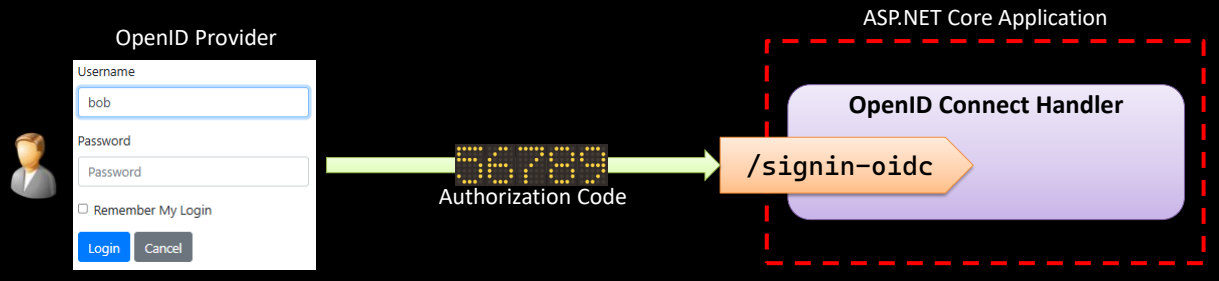
40

The Missing Piece

Handle the Authentication Callback



What Happens When It Receives the Code?

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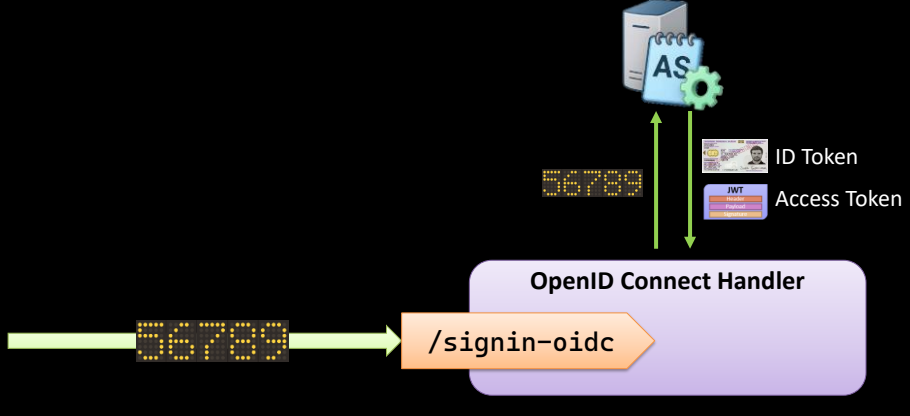
41

The Missing Piece

First, It Exchanges the Code for Tokens



What Happens Next?

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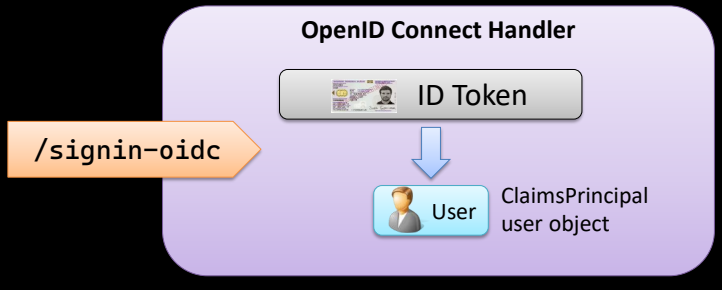
42

The Missing Piece

Next, It Creates a ClaimsPrincipal



What Does It Do With the User?

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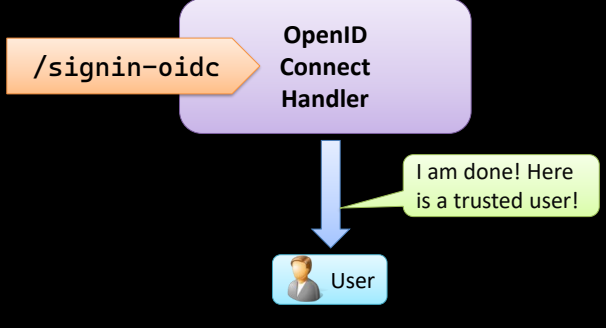
43

The Missing Piece

It Needs to Hand Off the User To Someone



Who Should It Give the User To?

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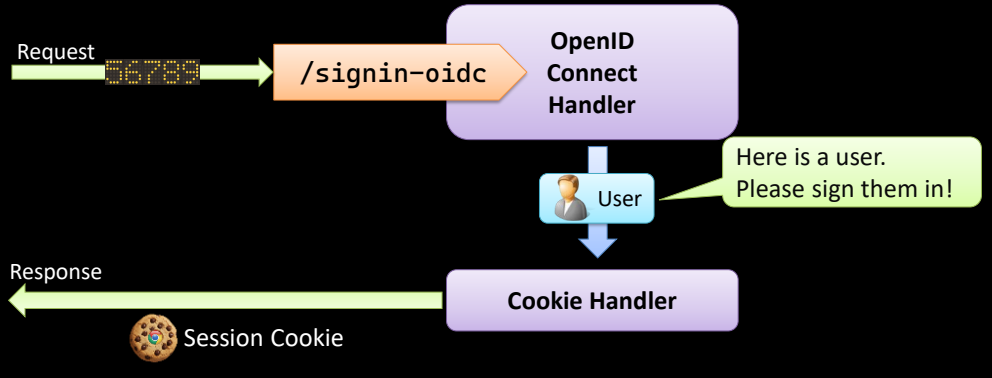
44

The Missing Piece

It Typically Gives It to the Cookie Handler



It Will Then Issue a Session Cookie

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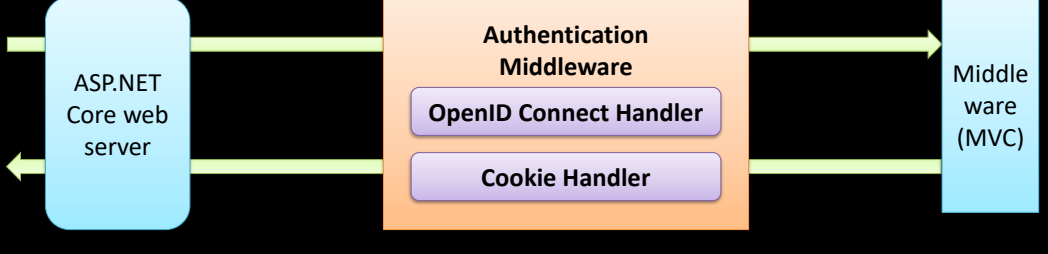
45

Live coding #5

```
builder.Services.AddAuthentication(o =>
{
    o.DefaultScheme = "cookie";
    o.DefaultChallengeScheme = "oidc";
})
.AddOpenIdConnect("oidc", o =>
{
   ...
}).AddCookie("cookie");
```



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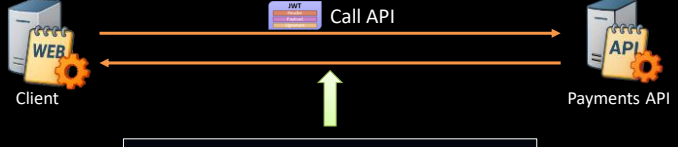
46

Using the Access Token

Using the Access Token

To Call the API We Include the Access Token

```
GET https://payment.secure.nu/payments
Authorization: Bearer eyJhbGciOiJSUzI1NiIsImt...
```




```
HTTP/1.1 200 OK
Content-Type: application/json

{ data }
```

Let Us Implement This!

Live coding #5

```
[HttpPost]
public async Task<IActionResult> GetPayments()
{
    var model = new PaymentApiModel();

    model.AccessToken = await HttpContext.GetTokenAsync("access_token") ?? "";

    var client = new HttpClient();
    client.DefaultRequestHeaders.Authorization =
        new AuthenticationHeaderValue("Bearer", model.AccessToken);

    var response = await client.GetAsync(PaymentsEndpoint);

    model.Result = await FormatResponseAsync(response);

    return View("Index", model);
}
```

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49

Where Is the Access Token?

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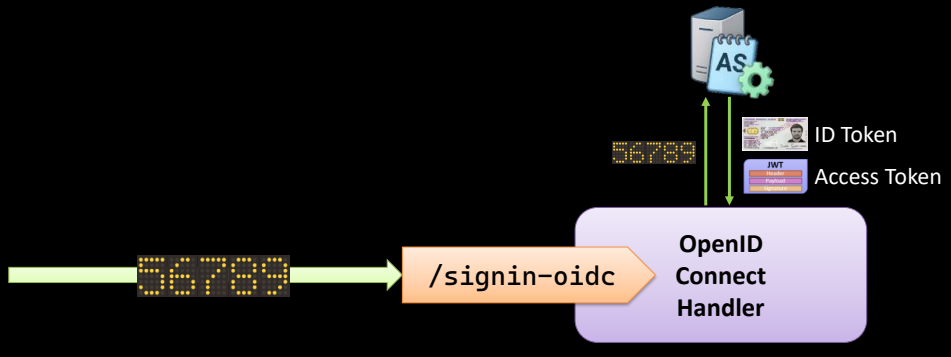
50

Where Is the Access Token?

The OpenID Connect Handler Received the Tokens



So, Where Did the Tokens Go?

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51

Where Is the Access Token?

By Default, the Tokens Are Not Saved

```
.AddOpenIdConnect("oidc", o =>
{
    ...
});
```

```
.AddOpenIdConnect("oidc", o =>
{
    ...
    o.SaveTokens = true;
});
```

What Happens When We Enable This?

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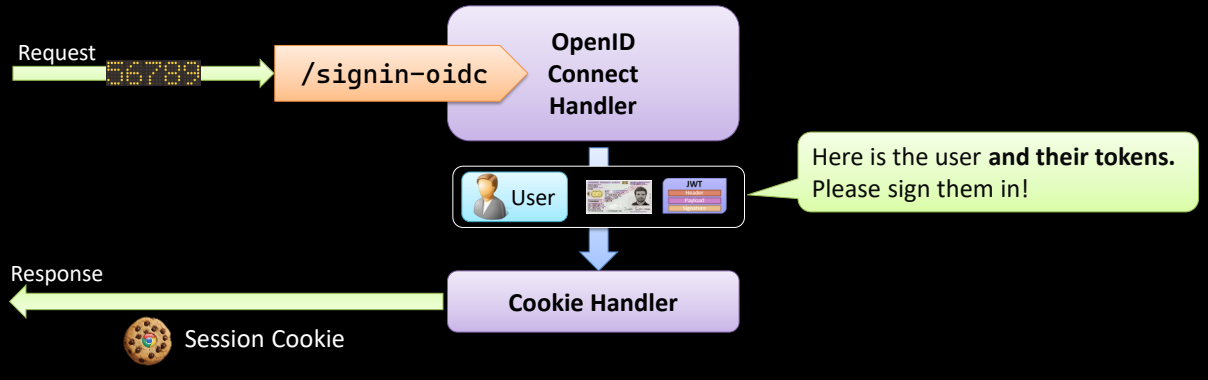
52

Where Is the Access Token?

It Passes the User and Tokens to the Cookie Handler



The Tokens Are Added to the Authentication Session

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53

Live coding #6

```
.AddOpenIdConnect("oidc", o =>
{
    ...
    o.SaveTokens = true;
});
```

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54

So, What Is OpenID Connect All About?

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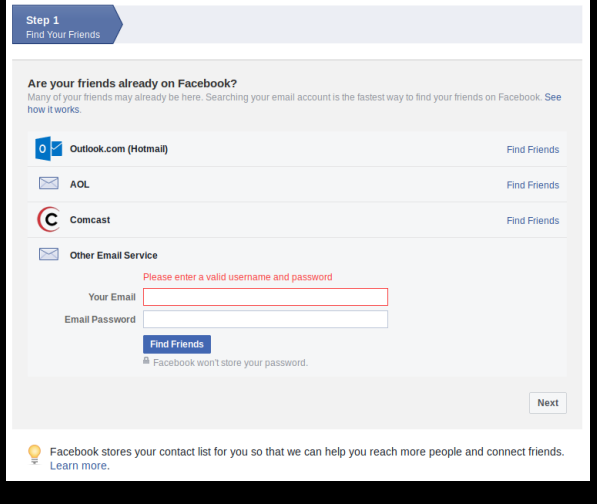
55

OpenID Connect

The Problem With Sharing Credentials



What Do We Want Instead?

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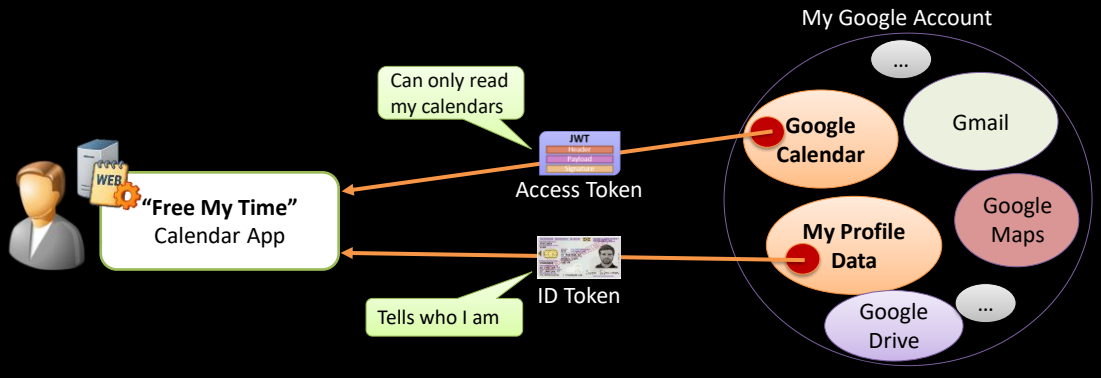
56

OpenID Connect

OpenID Connect Gives the Client What It Needs



- I Never Share My Google Password
- I Can Revoke Access at Any Time

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57

Questions?

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58